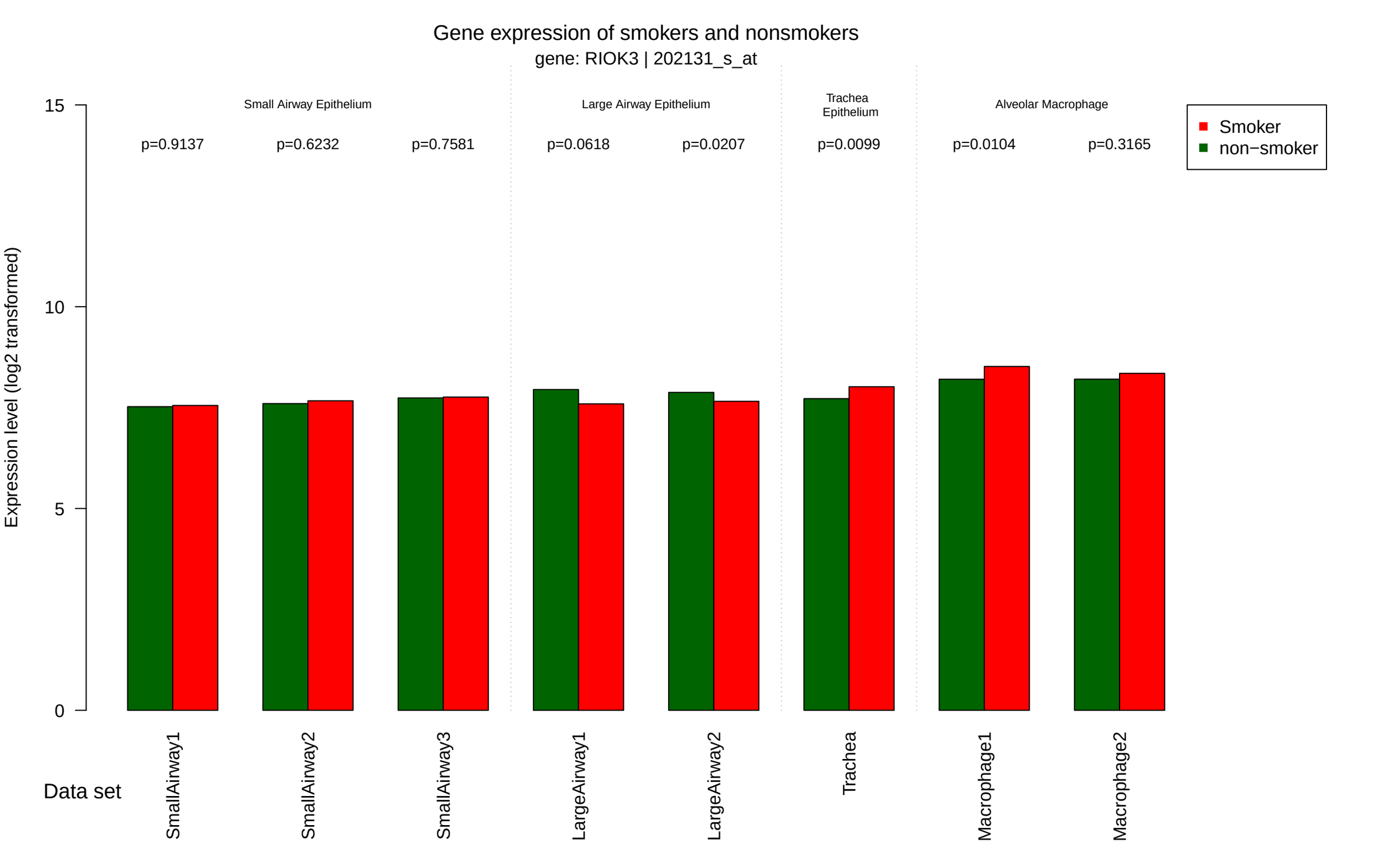
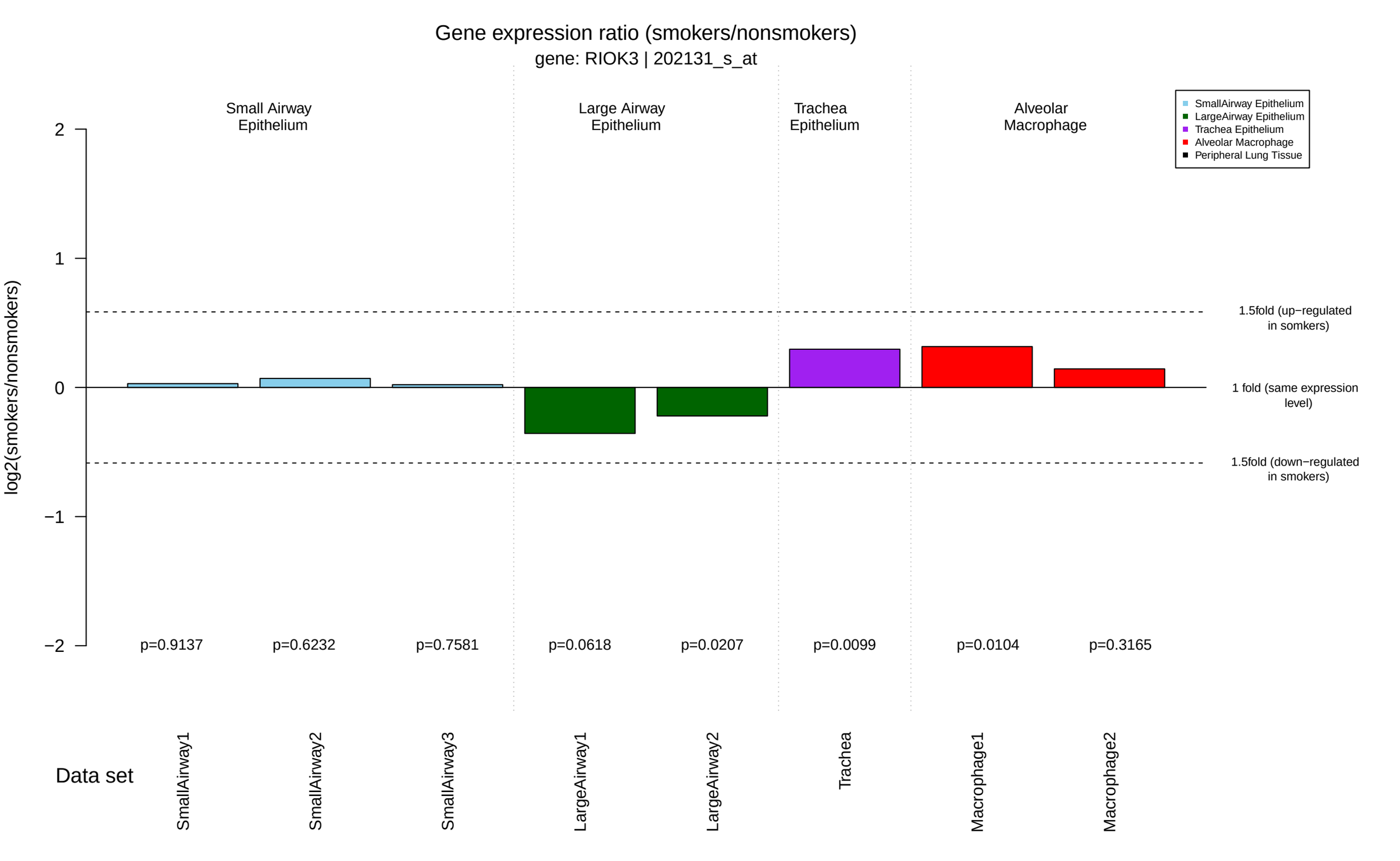
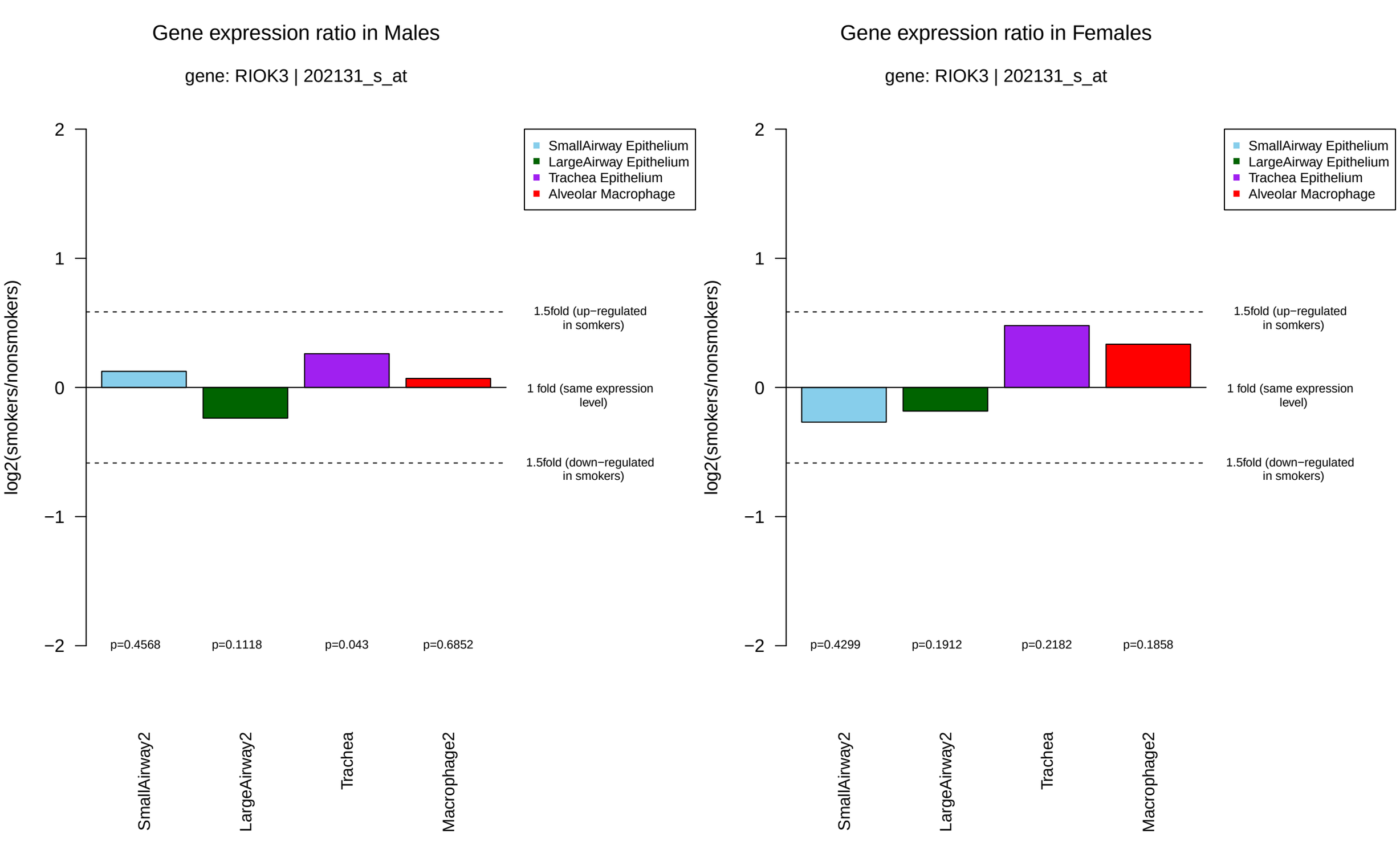
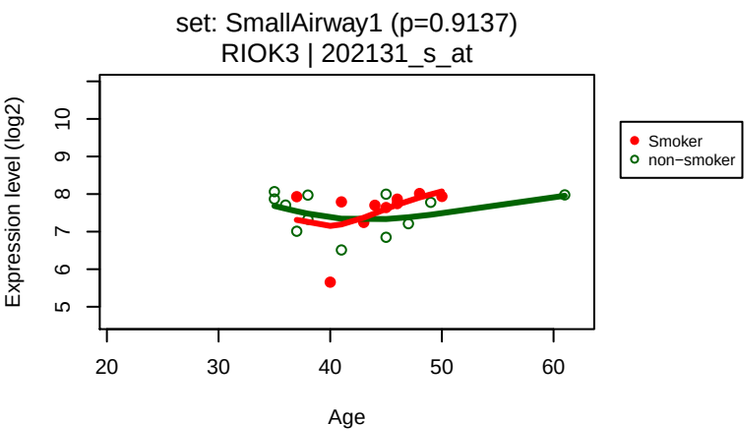
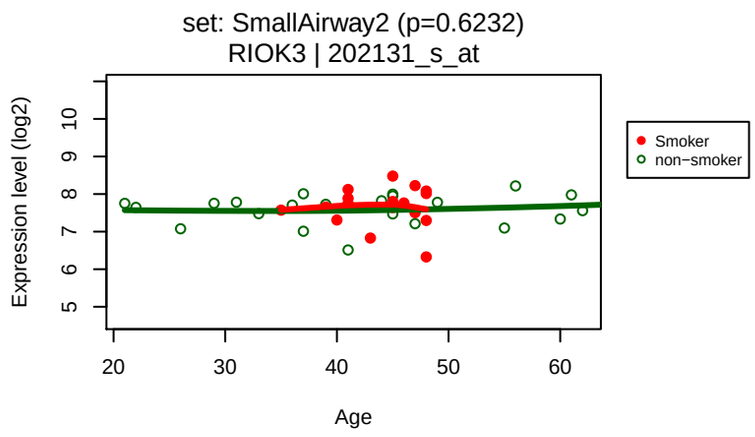
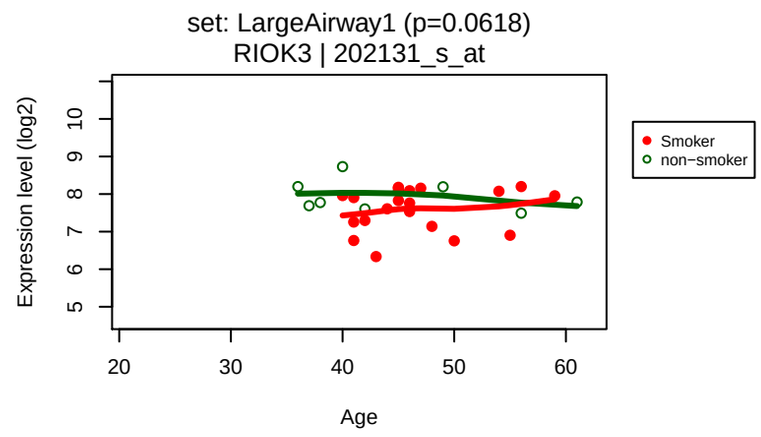

Gene expression of smokers and nonsmokers
gene: RIOK3 | 202131_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9137
p=0.6232
p=0.7581
p=0.0618
p=0.0207
p=0.0099
p=0.0104
p=0.3165
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

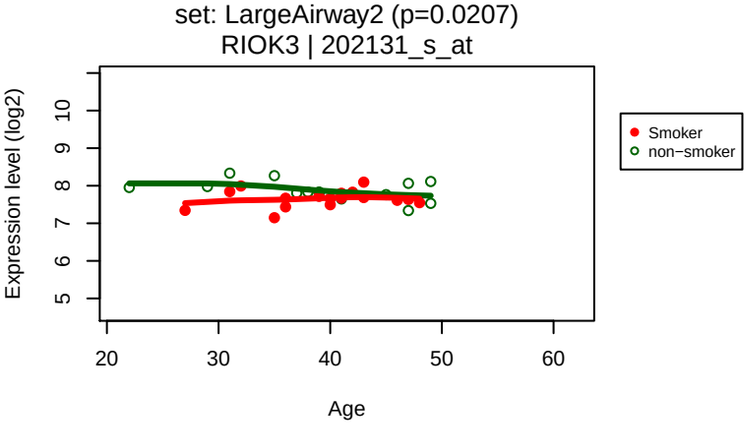

Gene expression ratio (smokers/nonsmokers)
gene: RIOK3 | 202131_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9137
p=0.6232
p=0.7581
p=0.0618
p=0.0207
p=0.0099
p=0.0104
p=0.3165
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

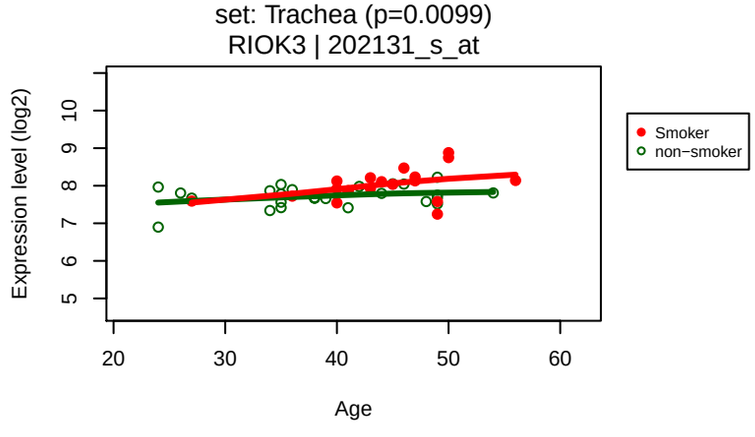

Gene expression ratio in Males
gene: RIOK3 | 202131_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4568
p=0.1118
p=0.043
p=0.6852
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RIOK3 | 202131_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4299
p=0.1912
p=0.2182
p=0.1858
SmallAirway2
LargeAirway2
Trachea
Macrophage2

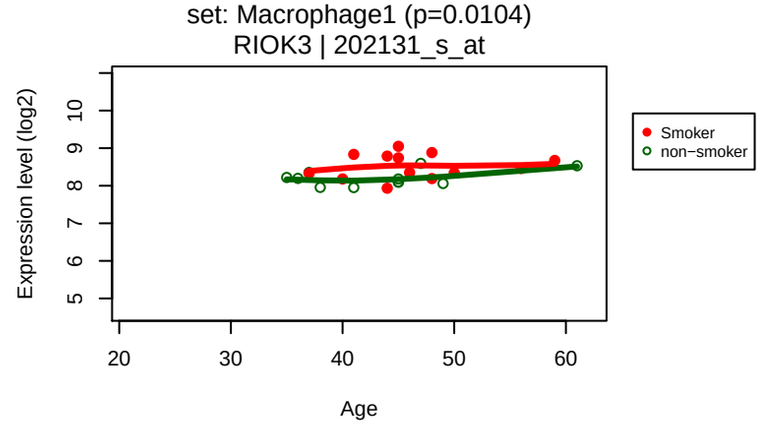

set: SmallAirway1 (p=0.9137)
RIOK3 | 202131_s_at
Expression level (log2)
Age
Smoker
non−smoker

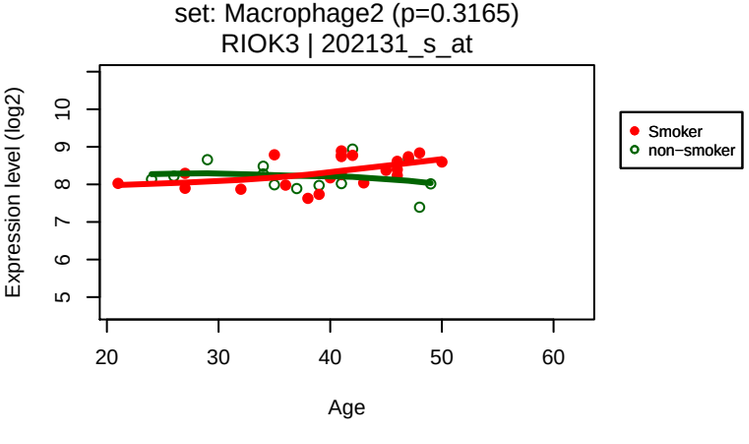

set: SmallAirway2 (p=0.6232)
RIOK3 | 202131_s_at
Expression level (log2)
Age
Smoker
non−smoker

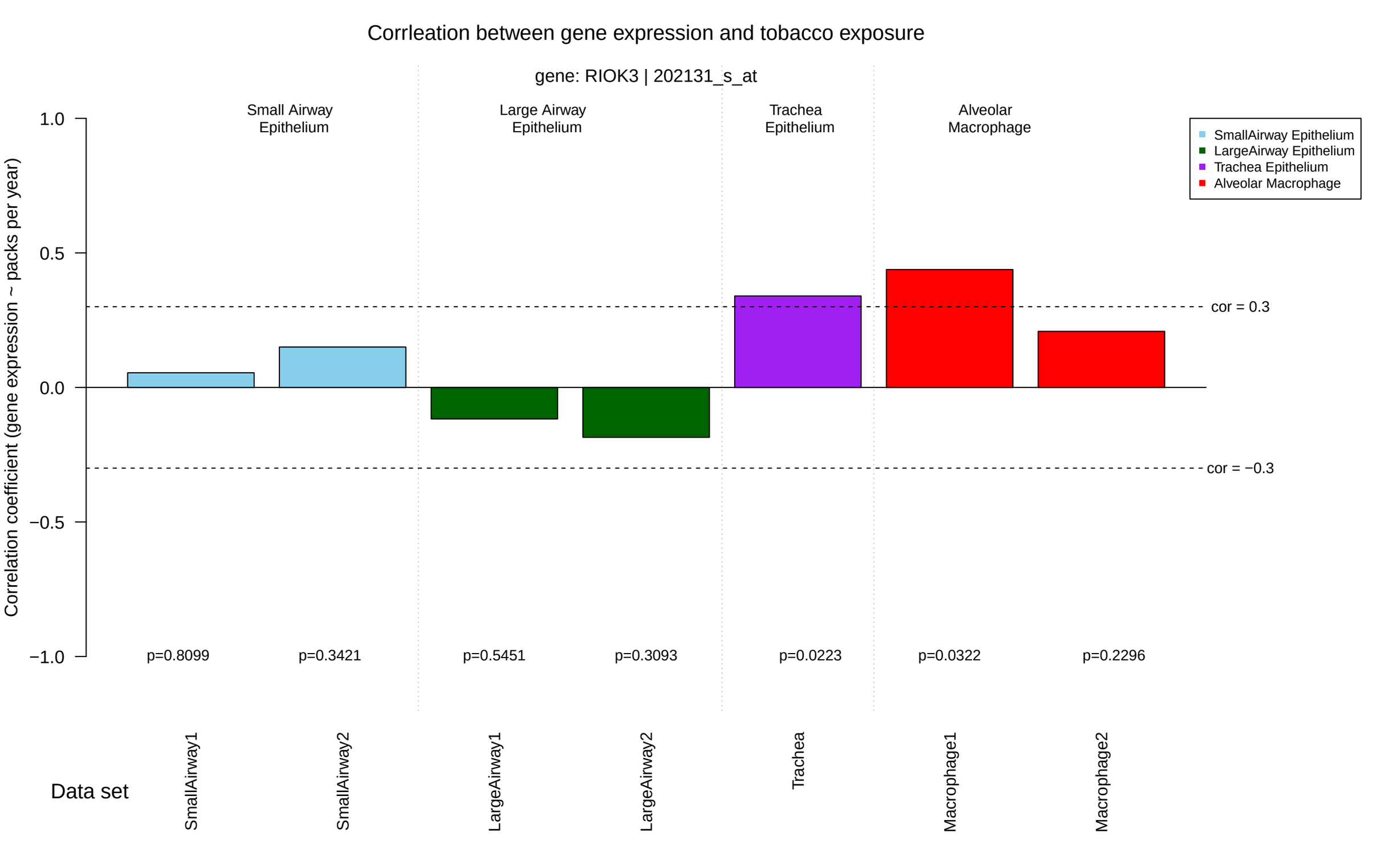

set: LargeAirway1 (p=0.0618)
RIOK3 | 202131_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0207)
RIOK3 | 202131_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0099)
RIOK3 | 202131_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0104)
RIOK3 | 202131_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3165)
RIOK3 | 202131_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RIOK3 | 202131_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8099
p=0.3421
p=0.5451
p=0.3093
p=0.0223
p=0.0322
p=0.2296
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2